

Reference excerpt 02-024

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on each sonogram of the video-clip The parameter ranges between 0 and 1 (typically 0.05) and serves to take account of heart rate variation (HR V) during acquisition. The t_{ai-1} is the instant corresponding to the wave a during the $(i - 1)$ -th cycle, the parameters t_{xi} and t_{vi} are dened similarly. The parameters t_{ac} , t_{ax} , t_{xv} and t_{vy} represent the time interval between the waves a and c, c and x, x and v, v and y, respectively (see Fig. 2) and are measured by an operator over a selected cardiac cycle. The detection algorithm takes into account their possible modification during subsequent cycles, by means of the parameter (see Eq. (1)). Once identified the waves a_i , c_i , x_i , v_i and y_i on each cardiac cycle, the intervals t_{xi} and t_{vi} are calculated for each cardiac cycle. The parameters dened in Eq. 1 are used to dene the two quantities $a_i = a_i - x_i$ $v_i = v_i - y_i$ (2) The average and the standard deviation for each parameter a_i , v_i , a_i , v_i , t_{xi} and t_{vi} are calculated over several cardiac cycle. An illustrative JVP trace is presented in Fig.3 where the waves a, c, x, v and y are visible. The parameters a_i , x_i , v_i and y_i for each cardiac cycle i were identified by the algorithm. However, about 10 % of the points are normally not properly detected by the algorithm, while it was possible to identify them manually (see Fig. 4). The parameters a_i , v_i , t_{xi} and t_{vi} were